

Maddula Lakshmi Sravya

Registration No.: 2310040032

Bachelor of Technology

Electronics and Communication Engineering

KLH University

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Electronics and Communication Engineering student with hands-on experience in FPGA-based digital design, RTL development using VHDL and Verilog, simulation, testbench development, functional verification, debugging, and Xilinx Vivado. Familiar with timing analysis, digital communication concepts, and hardware validation. Seeking opportunities in VHDL/FPGA RTL design and high-performance digital system development.

EDUCATION

• Koneru Lakshmaiah Education Foundation

2023-present

Bachelor Of Technology Electronics and Communication Engineering

CGPA: 9.86

• Sri Chaitanya Junior College

2021-2023

12th, State Board of Secondary Education

Percentage: 98%

• Lotus School

2020-2021

10th, State Board of Secondary Education

Percentage: 100%

PROJECTS

• FPGA-Based Defibrillator Pulse Detection Using ECG Signals

Designed an FPGA-based digital signal processing system for real-time ECG and defibrillator pulse detection.

- Developed RTL-based signal processing functionality and validated the design through simulation and hardware testing.
- Applied digital filtering to reduce ECG noise by 35–40%.
- Achieved 95% detection accuracy with real-time response below 10 ms.

• Transistor-Level Timing Analysis and Optimization of USB TX FIFO

Designed and verified USB TX FIFO at transistor level

- Evaluated key timing parameters: **propagation delay, setup time, and signal transitions**
- Optimized design achieving **9% latency reduction** and improved timing reliability
- Conducted simulation-based verification to improve design reliability

• Low-Power 3-Bit Encoder Using Memristor

Designed a 3-bit encoder using memristor-based logic with a focus on low-power operation.

- Evaluated parameters including power consumption, delay, and functional correctness.
- Simulated and verified circuit functionality and output behavior.

TECHNICAL SKILLS

Programming Languages: C, C++, Python, MATLAB, Java

HDL & RTL Design: VHDL, Verilog, SystemVerilog, RTL Design, RTL Simulation, Testbench Development, Functional Verification, UART, SPI

FPGA & Tools: FPGA Design, Xilinx Vivado, RTL Simulation, Synthesis, Timing Analysis, Waveform Debugging

Concepts: Digital Electronics, FPGA Architecture, ASIC Basics, Signal Processing, Embedded Systems

Verification: UVM, Test Case Development, Drivers, Monitors, Scoreboards, Debugging

Tools & Platforms: Cadence Virtuoso, Xilinx Vivado, MATLAB, Git, Linux

INTERSHIPS

• Taras Systems and Solutions — Design Verification Engineering Intern

- Implemented drivers, monitors, and scoreboards for validating design functionality.
- Performed testing, debugging, waveform analysis, and functional validation to identify design issues.

SOFT SKILLS

Communication, Problem Solving, Adaptability, Team Collaboration

CERTIFICATIONS

- VLSI Design and verification
- Advanced Automation Professional